SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Case Study 1: Video Conferencing Performance Issues

1.Analyze whether the problem is caused by the transport protocol or application layer.

The problem is **primarily caused by both the UDP transport protocol and the application layer buffering strategy**, but the **application layer is the main contributor**.

- Video conferencing applications generally use **UDP** because it provides low latency.
- UDP does not retransmit lost packets.
- An **8% packet loss** causes missing voice and video data.
- **180 ms jitter** causes packets to arrive at irregular intervals.
- If the application's jitter buffer is too small or poorly configured, it cannot compensate for these variations, resulting in voice distortion and frozen video.

2. Justification

Real-time multimedia applications require:

- Low latency
- Low jitter
- Continuous packet delivery
- Minimal retransmissions

TCP is unsuitable because:

- Retransmissions increase delay.
- Congestion control slows transmission.

UDP is preferred because:

- Faster transmission
- No retransmission delay
- Better for live audio and video

Hence, the application layer must effectively handle packet loss and jitter.

3.Recommendations

Protocol-Level Improvements

- Use **RTP over UDP**
- Enable **Forward Error Correction (FEC)**
- Apply **Quality of Service (QoS)**
- Prioritize multimedia traffic

Application-Level Improvements

- Increase adaptive jitter buffer size
- Implement adaptive bitrate streaming
- Use better codecs such as **Opus** and **H.265**
- Apply packet loss concealment techniques

Case Study 2: DNS Lookup Delay

1.Reasons for High DNS Resolution Time

Possible reasons include:

- Centralized DNS server
- Long network distance
- High DNS server load
- Multiple recursive queries
- Low cache hit rate
- Poor TTL configuration
- Network congestion

These factors increase DNS lookup to nearly **800 ms**.

2. Proposed DNS Architecture

Recommended Architecture

Client
↓
Local DNS Resolver
↓
Nearest Anycast DNS Server
↓
Authoritative DNS Servers

3.Techniques

- Deploy **Anycast DNS**
- Enable **DNS pre-fetching**
- Optimize TTL values (300–600 seconds)
- Use DNS caching
- Multiple geographically distributed DNS servers

Target lookup time:
< 50 ms

4.Advantages and Limitations

Advantages

- Faster response time
- High availability
- Better scalability
- Reduced network congestion
- Automatic failover

Limitations

- Higher deployment cost
- Complex management
- Cache inconsistency with high TTL
- Requires global Anycast infrastructure

Case Study 3: Centralized DNS Architecture

1.Impact of Centralized DNS

Problems include:

- Single point of failure
- Increased lookup latency
- Server overload
- Poor scalability
- Higher response time
- Reduced application performance

Users experience slow website loading during heavy traffic.

2.Distributed DNS Solution

Proposed Design

Users
↓
Nearest Anycast DNS Node
↓
Regional DNS Servers
↓
Primary & Secondary Authoritative DNS Servers

Features:

- Anycast routing
- Multiple regional DNS servers
- Recursive caching
- Load balancing
- Automatic failover

This reduces lookup latency significantly.

3.Effect of TTL, Anycast and DNS Cache

TTL

- High TTL
  - Fewer DNS queries
  - Faster response
  - Slower update propagation
- Low TTL
  - Faster updates
  - More DNS traffic

Anycast Routing

- Routes users to nearest DNS server
- Reduces latency
- Improves availability
- Balances load

DNS Cache

- Stores recent DNS records
- Reduces lookup delay
- Decreases server load
- Improves overall performance

Case Study 4: Stock Trading Platform

1.TCP Factors Responsible for Latency

(i) Flow Control

- Receiver advertises smaller window.
- Sender slows transmission.
- Order acknowledgments become delayed.

(ii) Nagle's Algorithm

- Combines small packets.
- Waits for acknowledgment.
- Adds delay to each trade request.

(iii) SYN Queue Limitation

- High connection requests overflow SYN queue.
- New TCP connections wait.
- Longer connection establishment time.

2. Effect on High-Frequency Trading

Flow Control

- Delays order processing
- Reduces throughput

Nagle's Algorithm

- Increases acknowledgment delay
- Slows order confirmation

SYN Queue Overflow

- Connection failures
- Increased waiting time
- Missed trading opportunities

Overall result:

Average latency increases from **1 ms to 40 ms**, affecting customer satisfaction.

4.TCP Configuration Recommendations

Disable Nagle's Algorithm

- Enable **TCP_NODELAY**
- Immediate packet transmission

Increase SYN Backlog Queue

- Handle more simultaneous connections
- Reduce connection delays

Window Scaling

- Improve flow control
- Increase throughput

Enable Selective Acknowledgment (SACK)

- Retransmit only lost packets
- Reduce unnecessary retransmissions

TCP Fast Open

- Reduce connection setup delay

QoS Configuration

- Prioritize trading packets